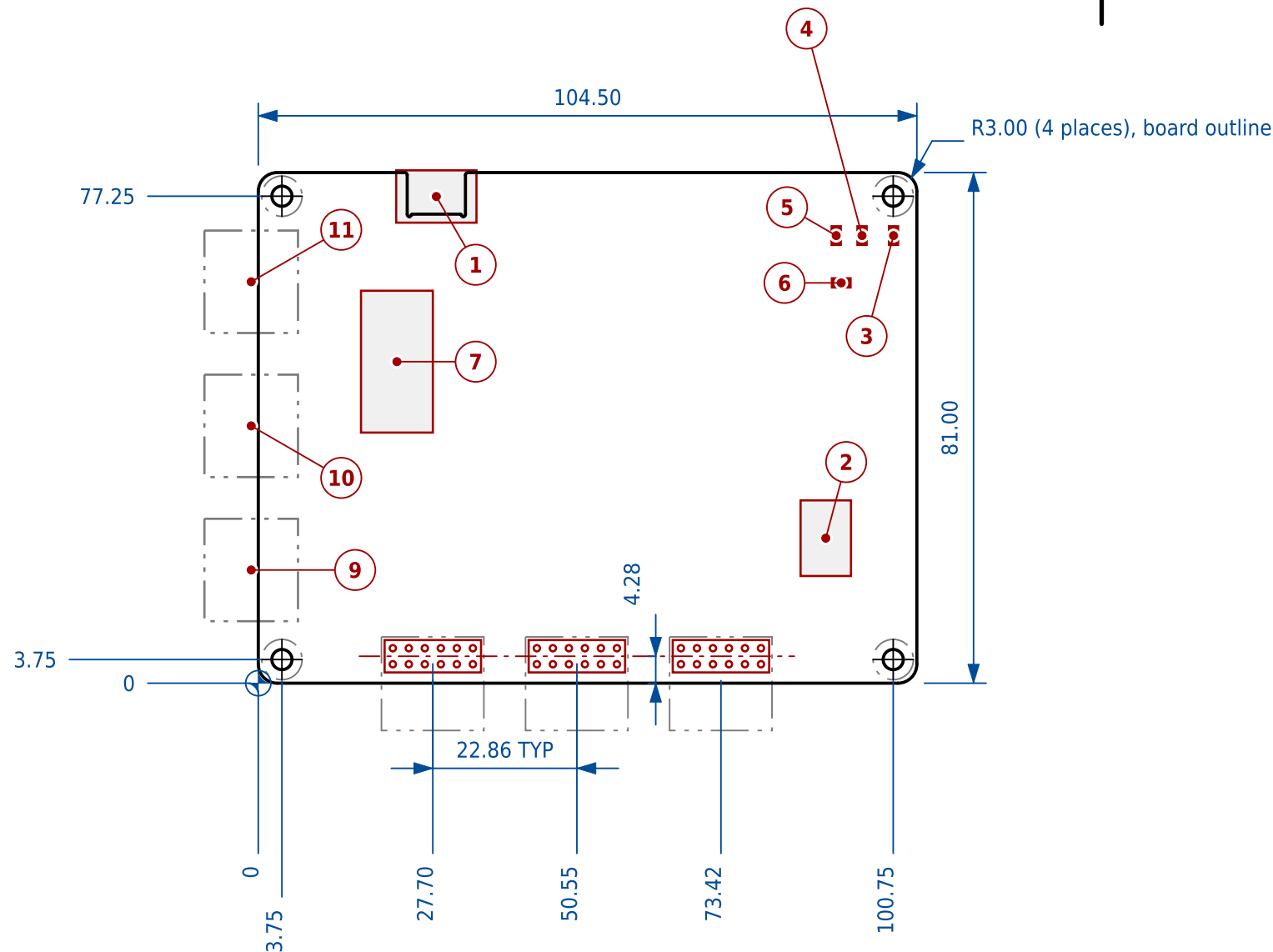




NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- v1.2.2c is an assembly variant, not a layout change: no X1, a 0R at R51. Its KiCad title block reads rev 1.2.3.
- This sheet covers revisions 1.2.2 and 1.2.3: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56252 mm is its stackup sum, not a specified finished thickness.

NOTES (continued)

- Assembled envelope, connector overhang included, is 104.50 x 81.35 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- Upper edge carries a shallow recess for the USB-C shell, contributed by the connector footprint's own edge cuts.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file: [@cfdd80d7b](https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb) - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26
- KiCad board file: [@a88cbc08b](https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb) - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11
- Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrfFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370 - used by: TT04, TT05

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.75	3.75	3.20	6.40
MT2	100.75	3.75	3.20	6.40
MT3	3.75	77.25	3.20	6.40
MT4	100.75	77.25	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	27.695	4.275	34.045	5.545
BIDIR	J5	bottom	50.555	4.275	56.905	5.545
OUTPUT	J6	bottom	73.415	4.275	79.765	5.545

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	21.88 to 34.62	73.05 to 81.35
2	7-segment display	86.05 to 94.05	17.00 to 29.00
3	LED D1	100.07 to 101.53	69.52 to 72.48
4	LED D2	95.07 to 96.53	69.52 to 72.48
5	LED D3	90.97 to 92.43	69.52 to 72.48
6	LED D4	91.02 to 93.98	62.77 to 64.23
7	8-way input DIP switch	16.30 to 27.70	39.75 to 62.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-8.55 to 6.30	9.83 to 26.08
10	Side Pmod J13 (not fitted)	-8.55 to 6.30	32.69 to 48.94
11	Side Pmod J14 (not fitted)	-8.55 to 6.30	55.55 to 71.80

LEGEND

—	Board outline
—	Component body, scheduled
- - -	Position not fitted on this revision
- - -	Pmod connector body, hole keep-out
—>	Dimension, extension and leader

DB 4+ V1.2.2 / DB 4+ V1.2.2C

tinytapeout-demo rev 1.2.2 and 1.2.3

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-88-g5151f5c+	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-02	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	